

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-01T10-44-04-356Z

Run ID: #2

Date Generated: 2026-01-01 10:44:09

Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming via induced pluripotency offers significant potential in regenerative medicine and cancer research. This experiment aimed to investigate the effects of repeated OSKM transfection on somatic cells under oncogenic stress conditions to evaluate the feasibility of cell fate reprogramming.

2. Materials and Methods

The experimental procedure involved six consecutive OSKM transfections on somatic cells subjected to oncogenic stress. Computational analysis was planned using single-cell variational inference (scVI) to assess cellular populations, though no analyses were conducted in this run.

Experimental Interventions:

- Oncogenic Stress Induced
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

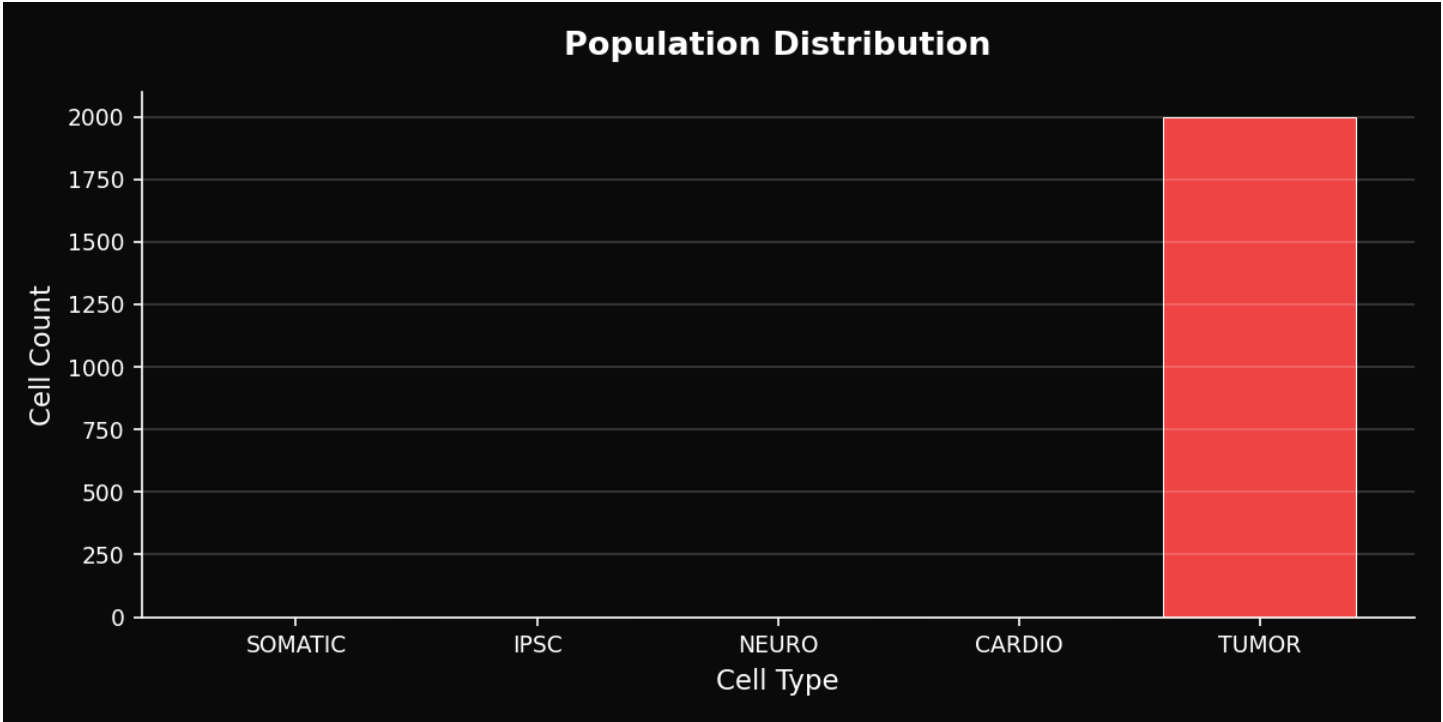
3. Results

The final cell population analysis revealed the absence of somatic, induced pluripotent stem cells (iPSCs), neural, and cardiac cell types, while exclusively yielding 2000 tumor cells. This outcome indicates a complete transformation of the initial somatic population towards tumorigenesis post-intervention.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	0	0.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	2000	100.0%

4. Discussion

The findings demonstrate that repeated OSKM transfection under oncogenic stress significantly favors tumorigenic conversion over pluripotency or differentiation into other cell types. This suggests a critical need to refine reprogramming protocols to prevent oncogenic outcomes. The absence of other cell types highlights potential limitations in the current approach and underscores the necessity for enhanced safeguards against tumor formation in reprogramming strategies.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.